	Scene	Atrium Day						Concourse Day					
	Camera	Pinhole	Fisheye1	Fisheye2	Fisheye3	360	Average	Pinhole	Fisheye1	Fisheye2	Fisheye3	360	Average
APR	PN [1]	15.5/85.5	11.8/79.9	11.3/80.4	10.1/79.8	4.5/82.6	10.6/81.6	18.6/95.0	10.7/88.7	10.1/87.7	7.7/85.9	2.6/39.0	9.9/79.3
APR	MS-T [1]	20.1/77.7	14.5/73.7	14.0/71.7	12.5/65.6	13.2/54.7	14.9/68.7	17.0/69.1	11.0/60.8	10.5/59.6	9.9/63.4	4.9/34.6	10.7/57.5
APR	PN-VC2 [1]	11.5/31.2	9.1/19.5	8.9/19.3	8.4/18.1	7.7/17.1	9.1/21.0	7.0/18.9	4.7/11.5	4.4/11.2	3.9/9.9	3.2/6.9	4.6/11.7
APR	MS-T-VC2 [1]	10.0/46.7	6.3/36.3	6.2/36.5	4.9/28.1	5.2/47.6	6.5/39.0	3.5/19.6	1.9/12.7	1.8/12.8	1.7/12.7	1.6/13.5	2.1/14.3
APR	CDG-Loc [2]	5.2/21.8	2.2/12.7	2.1/12.3	1.8/11.5	1.6/14.1	2.6/14.5	2.7/9.7	1.5/5.7	1.3/5.5	1.1/5.4	1.0/5.7	1.5/6.4
SCR	Ray-ACEG (Ours)	**1.9/4.3**	**1.2/2.9**	**1.2/3.0**	**1.1/2.7**	**0.9/2.3**	**1.3/3.0**	**1.4/6.0**	**0.9/4.4**	**1.0/4.5**	**0.8/4.1**	**0.7/3.5**	**1.0/4.5**
	Scene	Hall Day						Piatrium Day					
	Camera	Pinhole	Fisheye1	Fisheye2	Fisheye3	360	Average	Pinhole	Fisheye1	Fisheye2	Fisheye3	360	Average
APR	PN [1]	19.8/90.8	11.1/95.7	10.4/95.9	7.5/92.3	2.7/97.3	10.1/96.1	21.3/88.0	13.8/86.5	13.0/87.1	10.9/84.4	5.7/69.8	12.9/83.2
APR	MS-T [1]	21.8/96.0	14.1/94.0	13.5/92.7	11.2/91.7	11.4/88.5	14.4/92.6	30.0/78.5	23.4/71.3	22.7/72.7	19.3/72.6	11.3/52.0	21.3/69.4
APR	PN-VC2 [1]	7.0/26.4	4.4/15.5	4.3/14.7	3.9/14.2	3.2/12.2	4.6/16.6	10.4/24.5	6.9/13.3	6.8/12.8	6.8/11.5	5.0/9.8	7.2/14.4
APR	MS-T-VC2 [1]	4.0/36.9	2.2/25.8	2.1/25.2	1.8/26.2	1.7/26.9	2.4/28.2	8.1/32.7	4.0/24.5	3.9/23.6	3.8/24.1	3.3/22.1	4.6/25.4
APR	CDG-Loc [2]	2.3/17.4	**1.0**/8.9	**1.0**/8.6	**0.9**/8.4	**0.7**/7.5	1.2/10.2	4.4/18.0	**2.5**/11.3	**2.4**/10.5	**2.4**/10.1	**2.3**/10.1	**2.8**/12.0
SCR	Ray-ACEG (Ours)	**1.5/6.8**	1.1/**4.4**	1.0/**4.2**	0.9/**4.4**	0.9/**4.1**	**1.1/4.8**	**4.2/8.5**	2.8/**5.1**	2.6/**4.9**	2.7/**5.2**	2.4/**5.7**	2.9/**5.9**